JavaScript is disabled on your browser.

compact1, compact2, compact3

java.lang

Class StrictMath

java.lang.Object
    java.lang.StrictMath

```
public final class StrictMath
extends Object
```

The class `StrictMath` contains methods for performing basic numeric operations such as the elementary exponential, logarithm, square root, and trigonometric functions.

To help ensure portability of Java programs, the definitions of some of the numeric functions in this package require that they produce the same results as certain published algorithms. These algorithms are available from the well-known network library `netlib` as the package "Freely Distributable Math Library," `fdlibm`. These algorithms, which are written in the C programming language, are then to be understood as executed with all floating-point operations following the rules of Java floating-point arithmetic.

The Java math library is defined with respect to `fdlibm` version 5.3. Where `fdlibm` provides more than one definition for a function (such as `acos`), use the "IEEE 754 core function" version (residing in a file whose name begins with the letter `e`). The methods which require `fdlibm` semantics are `sin`, `cos`, `tan`, `asin`, `acos`, `atan`, `exp`, `log`, `log10`, `cbrt`, `atan2`, `pow`, `sinh`, `cosh`, `tanh`, `hypot`, `expm1`, and `log1p`.

The platform uses signed two's complement integer arithmetic with int and long primitive types. The developer should choose the primitive type to ensure that arithmetic operations consistently produce correct results, which in some cases means the operations will not overflow the range of values of the computation. The best practice is to choose the primitive type and algorithm to avoid overflow. In cases where the size is `int` or `long` and overflow errors need to be detected, the methods `addExact`, `subtractExact`, `multiplyExact`, and `toIntExact` throw an `ArithmeticException` when the results overflow. For other arithmetic operations

such as divide, absolute value, increment, decrement, and negation overflow occurs only with a specific minimum or maximum value and should be checked against the minimum or maximum as appropriate.

Since:

1.3

Field Summary

Fields

Modifier and Type	Field	Description
`static double`	`E`	The `double` value that is closer than any other to *e*, the base of the natural logarithms.
`static double`	`PI`	The `double` value that is closer than any other to *pi*, the ratio of the circumference of a circle to its diameter.

Method Summary

Concrete Methods

Modifier and Type	Method	Description
`static double`	`abs(double a)`	Returns the absolute value of a `double` value.
`static float`	`abs(float a)`	Returns the absolute value of a `float` value.
`static int`	`abs(int a)`	Returns the absolute value of an `int` value.
`static long`	`abs(long a)`	Returns the absolute value of a `long` value.
`static double`	`acos(double a)`	Returns the arc cosine of a value; the returned angle is in the range 0.0 through *pi*.
`static int`	`addExact(int x, int y)`	Returns the sum of its arguments, throwing an

		exception if the result overflows an `int`.
`static long`	**`addExact`**`(long x, long y)`	Returns the sum of its arguments, throwing an exception if the result overflows a `long`.
`static double`	**`asin`**`(double a)`	Returns the arc sine of a value; the returned angle is in the range -*pi*/2 through *pi*/2.
`static double`	**`atan`**`(double a)`	Returns the arc tangent of a value; the returned angle is in the range -*pi*/2 through *pi*/2.
`static double`	**`atan2`**`(double y, double x)`	Returns the angle *theta* from the conversion of rectangular coordinates (x, y) to polar coordinates (r, *theta*).
`static double`	**`cbrt`**`(double a)`	Returns the cube root of a `double` value.
`static double`	**`ceil`**`(double a)`	Returns the smallest (closest to negative infinity) `double` value that is greater than or equal to the argument and is equal to a mathematical integer.
`static double`	**`copySign`**`(double magnitude, double sign)`	Returns the first floating-point argument with the sign of the second floating-point argument.
`static float`	**`copySign`**`(float magnitude, float sign)`	Returns the first floating-point argument with the sign of the second floating-point argument.

static double	**cos**(double a)	Returns the trigonometric cosine of an angle.
static double	**cosh**(double x)	Returns the hyperbolic cosine of a `double` value.
static double	**exp**(double a)	Returns Euler's number *e* raised to the power of a `double` value.
static double	**expm1**(double x)	Returns e^x -1.
static double	**floor**(double a)	Returns the largest (closest to positive infinity) `double` value that is less than or equal to the argument and is equal to a mathematical integer.
static int	**floorDiv**(int x, int y)	Returns the largest (closest to positive infinity) `int` value that is less than or equal to the algebraic quotient.
static long	**floorDiv**(long x, long y)	Returns the largest (closest to positive infinity) `long` value that is less than or equal to the algebraic quotient.
static int	**floorMod**(int x, int y)	Returns the floor modulus of the `int` arguments.
static long	**floorMod**(long x, long y)	Returns the floor modulus of the `long` arguments.
static int	**getExponent**(double d)	Returns the unbiased exponent used in the representation of a `double`.
static int	**getExponent**(float f)	

		Returns the unbiased exponent used in the representation of a `float`.
`static double`	**`hypot`**`(double x, double y)`	Returns sqrt(x^2 +y^2) without intermediate overflow or underflow.
`static double`	**`IEEEremainder`**`(double f1, double f2)`	Computes the remainder operation on two arguments as prescribed by the IEEE 754 standard.
`static double`	**`log`**`(double a)`	Returns the natural logarithm (base e) of a `double` value.
`static double`	**`log10`**`(double a)`	Returns the base 10 logarithm of a `double` value.
`static double`	**`log1p`**`(double x)`	Returns the natural logarithm of the sum of the argument and 1.
`static double`	**`max`**`(double a, double b)`	Returns the greater of two `double` values.
`static float`	**`max`**`(float a, float b)`	Returns the greater of two `float` values.
`static int`	**`max`**`(int a, int b)`	Returns the greater of two `int` values.
`static long`	**`max`**`(long a, long b)`	Returns the greater of two `long` values.
`static double`	**`min`**`(double a, double b)`	Returns the smaller of two `double` values.
`static float`	**`min`**`(float a, float b)`	Returns the smaller of two `float` values.
`static int`	**`min`**`(int a, int b)`	Returns the smaller of two `int` values.
`static long`	**`min`**`(long a, long b)`	Returns the smaller of two `long` values.

static int	**multiplyExact**(int x, int y)	Returns the product of the arguments, throwing an exception if the result overflows an int.
static long	**multiplyExact**(long x, long y)	Returns the product of the arguments, throwing an exception if the result overflows a long.
static double	**nextAfter**(double start, double direction)	Returns the floating-point number adjacent to the first argument in the direction of the second argument.
static float	**nextAfter**(float start, double direction)	Returns the floating-point number adjacent to the first argument in the direction of the second argument.
static double	**nextDown**(double d)	Returns the floating-point value adjacent to d in the direction of negative infinity.
static float	**nextDown**(float f)	Returns the floating-point value adjacent to f in the direction of negative infinity.
static double	**nextUp**(double d)	Returns the floating-point value adjacent to d in the direction of positive infinity.
static float	**nextUp**(float f)	Returns the floating-point value adjacent to f in the direction of positive infinity.
static double	**pow**(double a, double b)	Returns the value of the first argument raised to the power of the second argument.

static double	**random**()	Returns a double value with a positive sign, greater than or equal to 0.0 and less than 1.0.
static double	**rint**(double a)	Returns the double value that is closest in value to the argument and is equal to a mathematical integer.
static long	**round**(double a)	Returns the closest long to the argument, with ties rounding to positive infinity.
static int	**round**(float a)	Returns the closest int to the argument, with ties rounding to positive infinity.
static double	**scalb**(double d, int scaleFactor)	Returns d × $2^{\texttt{scaleFactor}}$ rounded as if performed by a single correctly rounded floating-point multiply to a member of the double value set.
static float	**scalb**(float f, int scaleFactor)	Returns f × $2^{\texttt{scaleFactor}}$ rounded as if performed by a single correctly rounded floating-point multiply to a member of the float value set.
static double	**signum**(double d)	Returns the signum function of the argument; zero if the argument is zero, 1.0 if the argument is greater than zero, -1.0 if the argument is less than zero.
static float	**signum**(float f)	Returns the signum function of the

		argument; zero if the argument is zero, 1.0f if the argument is greater than zero, -1.0f if the argument is less than zero.
static double	**sin**(double a)	Returns the trigonometric sine of an angle.
static double	**sinh**(double x)	Returns the hyperbolic sine of a double value.
static double	**sqrt**(double a)	Returns the correctly rounded positive square root of a double value.
static int	**subtractExact**(int x, int y)	Returns the difference of the arguments, throwing an exception if the result overflows an int.
static long	**subtractExact**(long x, long y)	Returns the difference of the arguments, throwing an exception if the result overflows a long.
static double	**tan**(double a)	Returns the trigonometric tangent of an angle.
static double	**tanh**(double x)	Returns the hyperbolic tangent of a double value.
static double	**toDegrees**(double angrad)	Converts an angle measured in radians to an approximately equivalent angle measured in degrees.
static int	**toIntExact**(long value)	Returns the value of the long argument; throwing an

		exception if the value overflows an `int`.
`static double`	**`toRadians`**`(double angdeg)`	Converts an angle measured in degrees to an approximately equivalent angle measured in radians.
`static double`	**`ulp`**`(double d)`	Returns the size of an ulp of the argument.
`static float`	**`ulp`**`(float f)`	Returns the size of an ulp of the argument.

Methods inherited from class java.lang.Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Field Detail

E

```
public static final double E
```

The `double` value that is closer than any other to *e*, the base of the natural logarithms.

See Also:

Constant Field Values

PI

```
public static final double PI
```

The `double` value that is closer than any other to *pi*, the ratio of the circumference of a circle to its diameter.

See Also:

Constant Field Values

Method Detail

sin

```
public static double sin(double a)
```

Returns the trigonometric sine of an angle. Special cases:

- If the argument is NaN or an infinity, then the result is NaN.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`a - an angle, in radians.`

Returns:

`the sine of the argument.`

cos

```
public static double cos(double a)
```

Returns the trigonometric cosine of an angle. Special cases:

- If the argument is NaN or an infinity, then the result is NaN.

Parameters:

`a - an angle, in radians.`

Returns:

`the cosine of the argument.`

tan

```
public static double tan(double a)
```

Returns the trigonometric tangent of an angle. Special cases:

- If the argument is NaN or an infinity, then the result is NaN.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`a - an angle, in radians.`

Returns:

`the tangent of the argument.`

asin

```
public static double asin(double a)
```

Returns the arc sine of a value; the returned angle is in the range -*pi*/2 through *pi*/2. Special cases:

- If the argument is NaN or its absolute value is greater than 1, then the result is NaN.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`a - the value whose arc sine is to be returned.`

Returns:

`the arc sine of the argument.`

acos

```
public static double acos(double a)
```

Returns the arc cosine of a value; the returned angle is in the range 0.0 through *pi*. Special case:

- If the argument is NaN or its absolute value is greater than 1, then the result is NaN.

Parameters:

`a - the value whose arc cosine is to be returned.`

Returns:

`the arc cosine of the argument.`

atan

```
public static double atan(double a)
```

Returns the arc tangent of a value; the returned angle is in the range -*pi*/2 through *pi*/2. Special cases:

- If the argument is NaN, then the result is NaN.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`a - the value whose arc tangent is to be returned.`

Returns:

`the arc tangent of the argument.`

toRadians

```
public static double toRadians(double angdeg)
```

Converts an angle measured in degrees to an approximately equivalent angle measured in radians. The conversion from degrees to radians is generally inexact.

Parameters:

`angdeg - an angle, in degrees`

Returns:

`the measurement of the angle angdeg in radians.`

toDegrees

```
public static double toDegrees(double angrad)
```

Converts an angle measured in radians to an approximately equivalent angle measured in degrees. The conversion from radians to degrees is generally inexact; users should *not* expect `cos(toRadians(90.0))` to exactly equal `0.0`.

Parameters:

`angrad - an angle, in radians`

Returns:

`the measurement of the angle angrad in degrees.`

exp

```
public static double exp(double a)
```

Returns Euler's number *e* raised to the power of a `double` value. Special cases:

- If the argument is NaN, the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is negative infinity, then the result is positive zero.

Parameters:

`a - the exponent to raise e to.`

Returns:

the value e^a, where *e* is the base of the natural logarithms.

log

```
public static double log(double a)
```

Returns the natural logarithm (base *e*) of a `double` value. Special cases:

- If the argument is NaN or less than zero, then the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is positive zero or negative zero, then the result is negative infinity.

Parameters:

`a - a value`

Returns:

`the value ln a, the natural logarithm of a.`

log10

```
public static double log10(double a)
```

Returns the base 10 logarithm of a `double` value. Special cases:

- If the argument is NaN or less than zero, then the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is positive zero or negative zero, then the result is negative infinity.
- If the argument is equal to 10^n for integer n, then the result is n.

Parameters:

`a - a value`

Returns:

`the base 10 logarithm of a.`

Since:

`1.5`

sqrt

```
public static double sqrt(double a)
```

Returns the correctly rounded positive square root of a `double` value. Special cases:

- If the argument is NaN or less than zero, then the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is positive zero or negative zero, then the result is the same as the argument.

Otherwise, the result is the `double` value closest to the true mathematical square root of the argument value.

Parameters:

`a - a value.`

Returns:

`the positive square root of a.`

cbrt

```
public static double cbrt(double a)
```

Returns the cube root of a `double` value. For positive finite `x`, `cbrt(-x) == -cbrt(x)`; that is, the cube root of a negative value is the negative of the cube root of that value's magnitude. Special cases:

- If the argument is NaN, then the result is NaN.
- If the argument is infinite, then the result is an infinity with the same sign as the argument.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`a - a value.`

Returns:

`the cube root of a.`

Since:

`1.5`

IEEEremainder

```
public static double IEEEremainder(double f1,
                                   double f2)
```

Computes the remainder operation on two arguments as prescribed by the IEEE 754 standard. The remainder value is mathematically equal to `f1 - f2` × *n*, where *n* is the mathematical integer closest to the exact mathematical value of the quotient `f1/f2`, and if two mathematical integers are equally close to `f1/f2`, then *n* is the integer that is even. If the remainder is zero, its sign is the same as the sign of the first argument. Special cases:

- If either argument is NaN, or the first argument is infinite, or the second argument is positive zero or negative zero, then the result is NaN.
- If the first argument is finite and the second argument is infinite, then the result is the same as the first argument.

Parameters:

`f1 - the dividend.`

`f2 - the divisor.`

Returns:

the remainder when f1 is divided by f2.

ceil

```
public static double ceil(double a)
```

Returns the smallest (closest to negative infinity) `double` value that is greater than or equal to the argument and is equal to a mathematical integer. Special cases:

- If the argument value is already equal to a mathematical integer, then the result is the same as the argument.
- If the argument is NaN or an infinity or positive zero or negative zero, then the result is the same as the argument.
- If the argument value is less than zero but greater than -1.0, then the result is negative zero.

Note that the value of `StrictMath.ceil(x)` is exactly the value of `-StrictMath.floor(-x)`.

Parameters:

a - a value.

Returns:

the smallest (closest to negative infinity) floating-point value that is greater than or equal to the argument and is equal to a mathematical integer.

floor

```
public static double floor(double a)
```

Returns the largest (closest to positive infinity) `double` value that is less than or equal to the argument and is equal to a mathematical integer. Special cases:

- If the argument value is already equal to a mathematical integer, then the result is the same as the argument.
- If the argument is NaN or an infinity or positive zero or negative zero, then the result is the same as the argument.

Parameters:

a - a value.

Returns:

the largest (closest to positive infinity) floating-point value that less than or equal to the argument and is equal to a mathematical integer.

rint

```
public static double rint(double a)
```

Returns the `double` value that is closest in value to the argument and is equal to a mathematical integer. If two `double` values that are mathematical integers are equally close to the value of the argument, the result is the integer value that is even. Special cases:

- If the argument value is already equal to a mathematical integer, then the result is the same as the argument.
- If the argument is NaN or an infinity or positive zero or negative zero, then the result is the same as the argument.

Parameters:

`a - a value.`

Returns:

`the closest floating-point value to a that is equal to a mathematical integer.`

atan2

```
public static double atan2(double y,
                           double x)
```

Returns the angle *theta* from the conversion of rectangular coordinates (`x`, `y`) to polar coordinates (r, *theta*). This method computes the phase *theta* by computing an arc tangent of `y/x` in the range of -*pi* to *pi*. Special cases:

- If either argument is NaN, then the result is NaN.
- If the first argument is positive zero and the second argument is positive, or the first argument is positive and finite and the second argument is positive infinity, then the result is positive zero.
- If the first argument is negative zero and the second argument is positive, or the first argument is negative and finite and the second argument is positive infinity, then the result is negative zero.
- If the first argument is positive zero and the second argument is negative, or the first argument is positive and finite and the second argument is negative infinity, then the result is the `double` value closest to *pi*.
- If the first argument is negative zero and the second argument is negative, or the first argument is negative and finite and the second argument is negative infinity, then the result is the `double` value closest to -*pi*.
- If the first argument is positive and the second argument is positive zero or negative zero, or the first argument is positive infinity and the second argument is finite, then the result is the `double` value closest to *pi*/2.

- If the first argument is negative and the second argument is positive zero or negative zero, or the first argument is negative infinity and the second argument is finite, then the result is the `double` value closest to -*pi*/2.
- If both arguments are positive infinity, then the result is the `double` value closest to *pi*/4.
- If the first argument is positive infinity and the second argument is negative infinity, then the result is the `double` value closest to 3**pi*/4.
- If the first argument is negative infinity and the second argument is positive infinity, then the result is the `double` value closest to -*pi*/4.
- If both arguments are negative infinity, then the result is the `double` value closest to -3**pi*/4.

Parameters:

`y - the ordinate coordinate`

`x - the abscissa coordinate`

Returns:

`the` *`theta`* `component of the point (`*`r`*`,` *`theta`*`) in polar coordinates that corresponds to the point (`*`x`*`,` *`y`*`) in Cartesian coordinates.`

pow

```
public static double pow(double a,
                         double b)
```

Returns the value of the first argument raised to the power of the second argument. Special cases:

- If the second argument is positive or negative zero, then the result is 1.0.
- If the second argument is 1.0, then the result is the same as the first argument.
- If the second argument is NaN, then the result is NaN.
- If the first argument is NaN and the second argument is nonzero, then the result is NaN.
- If
    - the absolute value of the first argument is greater than 1 and the second argument is positive infinity, or
    - the absolute value of the first argument is less than 1 and the second argument is negative infinity,

  then the result is positive infinity.
- If
    - the absolute value of the first argument is greater than 1 and the second argument is negative infinity, or
    - the absolute value of the first argument is less than 1 and the second argument is positive infinity,

  then the result is positive zero.

- If the absolute value of the first argument equals 1 and the second argument is infinite, then the result is NaN.
- If
    - the first argument is positive zero and the second argument is greater than zero, or
    - the first argument is positive infinity and the second argument is less than zero,

  then the result is positive zero.
- If
    - the first argument is positive zero and the second argument is less than zero, or
    - the first argument is positive infinity and the second argument is greater than zero,

  then the result is positive infinity.
- If
    - the first argument is negative zero and the second argument is greater than zero but not a finite odd integer, or
    - the first argument is negative infinity and the second argument is less than zero but not a finite odd integer,

  then the result is positive zero.
- If
    - the first argument is negative zero and the second argument is a positive finite odd integer, or
    - the first argument is negative infinity and the second argument is a negative finite odd integer,

  then the result is negative zero.
- If
    - the first argument is negative zero and the second argument is less than zero but not a finite odd integer, or
    - the first argument is negative infinity and the second argument is greater than zero but not a finite odd integer,

  then the result is positive infinity.
- If
    - the first argument is negative zero and the second argument is a negative finite odd integer, or
    - the first argument is negative infinity and the second argument is a positive finite odd integer,

  then the result is negative infinity.
- If the first argument is finite and less than zero
    - if the second argument is a finite even integer, the result is equal to the result of raising the absolute value of the first argument to the power of the second argument
    - if the second argument is a finite odd integer, the result is equal to the negative of the result of raising the absolute value of the first argument to the power of the second argument
    - if the second argument is finite and not an integer, then the result is NaN.
- If both arguments are integers, then the result is exactly equal to the mathematical result of raising the first argument to the power of the

second argument if that result can in fact be represented exactly as a `double` value.

(In the foregoing descriptions, a floating-point value is considered to be an integer if and only if it is finite and a fixed point of the method `ceil` or, equivalently, a fixed point of the method `floor`. A value is a fixed point of a one-argument method if and only if the result of applying the method to the value is equal to the value.)

Parameters:

`a - base.`

`b - the exponent.`

Returns:

`the value a`^{b}`.`

round

`public static int round(float a)`

Returns the closest `int` to the argument, with ties rounding to positive infinity.

Special cases:

- If the argument is NaN, the result is 0.
- If the argument is negative infinity or any value less than or equal to the value of `Integer.MIN_VALUE`, the result is equal to the value of `Integer.MIN_VALUE`.
- If the argument is positive infinity or any value greater than or equal to the value of `Integer.MAX_VALUE`, the result is equal to the value of `Integer.MAX_VALUE`.

Parameters:

`a - a floating-point value to be rounded to an integer.`

Returns:

`the value of the argument rounded to the nearest int value.`

See Also:

`Integer.MAX_VALUE`, `Integer.MIN_VALUE`

round

`public static long round(double a)`

Returns the closest `long` to the argument, with ties rounding to positive infinity.

Special cases:

- If the argument is NaN, the result is 0.
- If the argument is negative infinity or any value less than or equal to the value of `Long.MIN_VALUE`, the result is equal to the value of `Long.MIN_VALUE`.
- If the argument is positive infinity or any value greater than or equal to the value of `Long.MAX_VALUE`, the result is equal to the value of `Long.MAX_VALUE`.

Parameters:

`a - a floating-point value to be rounded to a long.`

Returns:

`the value of the argument rounded to the nearest long value.`

See Also:

`Long.MAX_VALUE`, `Long.MIN_VALUE`

random

```
public static double random()
```

Returns a `double` value with a positive sign, greater than or equal to `0.0` and less than `1.0`. Returned values are chosen pseudorandomly with (approximately) uniform distribution from that range.

When this method is first called, it creates a single new pseudorandom-number generator, exactly as if by the expression

```
new java.util.Random()
```

This new pseudorandom-number generator is used thereafter for all calls to this method and is used nowhere else.

This method is properly synchronized to allow correct use by more than one thread. However, if many threads need to generate pseudorandom numbers at a great rate, it may reduce contention for each thread to have its own pseudorandom-number generator.

Returns:

`a pseudorandom double greater than or equal to 0.0 and less than 1.0.`

See Also:

`Random.nextDouble()`

addExact

```
public static int addExact(int x,
                           int y)
```

Returns the sum of its arguments, throwing an exception if the result overflows an `int`.

Parameters:

`x - the first value`

`y - the second value`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows an int`

Since:

`1.8`

See Also:

`Math.addExact(int,int)`

addExact

```
public static long addExact(long x,
                            long y)
```

Returns the sum of its arguments, throwing an exception if the result overflows a `long`.

Parameters:

`x - the first value`

`y - the second value`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows a long`

Since:

`1.8`

See Also:

`Math.addExact(long,long)`

subtractExact

```
public static int subtractExact(int x,
                                int y)
```

Returns the difference of the arguments, throwing an exception if the result overflows an `int`.

Parameters:

`x - the first value`

`y - the second value to subtract from the first`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows an int`

Since:

`1.8`

See Also:

`Math.subtractExact(int,int)`

subtractExact

```
public static long subtractExact(long x,
                                 long y)
```

Returns the difference of the arguments, throwing an exception if the result overflows a `long`.

Parameters:

`x - the first value`

`y - the second value to subtract from the first`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows a long`

Since:

`1.8`

See Also:

`Math.subtractExact(long,long)`

multiplyExact

```
public static int multiplyExact(int x,
                                int y)
```

Returns the product of the arguments, throwing an exception if the result overflows an `int`.

Parameters:

`x - the first value`

`y - the second value`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows an int`

Since:

`1.8`

See Also:

`Math.multiplyExact(int,int)`

multiplyExact

```
public static long multiplyExact(long x,
                                 long y)
```

Returns the product of the arguments, throwing an exception if the result overflows a `long`.

Parameters:

`x - the first value`

`y - the second value`

Returns:

`the result`

Throws:

`ArithmeticException - if the result overflows a long`

Since:

`1.8`

See Also:

`Math.multiplyExact(long,long)`

toIntExact

```
public static int toIntExact(long value)
```

Returns the value of the `long` argument; throwing an exception if the value overflows an `int`.

Parameters:

`value - the long value`

Returns:

`the argument as an int`

Throws:

`ArithmeticException - if the argument overflows an int`

Since:

`1.8`

See Also:

`Math.toIntExact(long)`

floorDiv

```
public static int floorDiv(int x,
                           int y)
```

Returns the largest (closest to positive infinity) `int` value that is less than or equal to the algebraic quotient. There is one special case, if the dividend is the Integer.MIN_VALUE and the divisor is `-1`, then integer overflow occurs and the result is equal to the `Integer.MIN_VALUE`.

See `Math.floorDiv` for examples and a comparison to the integer division `/` operator.

Parameters:

`x - the dividend`

`y - the divisor`

Returns:

`the largest (closest to positive infinity) int value that is less than or equal to the algebraic quotient.`

Throws:

`ArithmeticException - if the divisor y is zero`

Since:

`1.8`

See Also:

`Math.floorDiv(int, int)`, `Math.floor(double)`

floorDiv

```
public static long floorDiv(long x,
                            long y)
```

Returns the largest (closest to positive infinity) `long` value that is less than or equal to the algebraic quotient. There is one special case, if the dividend is the Long.MIN_VALUE and the divisor is `-1`, then integer overflow occurs and the result is equal to the `Long.MIN_VALUE`.

See `Math.floorDiv` for examples and a comparison to the integer division `/` operator.

Parameters:

`x - the dividend`

`y - the divisor`

Returns:

the largest (closest to positive infinity) long value that is less than or equal to the algebraic quotient.

Throws:

`ArithmeticException` - if the divisor `y` is zero

Since:

1.8

See Also:

`Math.floorDiv(long, long)`, `Math.floor(double)`

floorMod

```
public static int floorMod(int x,
                           int y)
```

Returns the floor modulus of the `int` arguments.

The floor modulus is `x - (floorDiv(x, y) * y)`, has the same sign as the divisor `y`, and is in the range of `-abs(y) < r < +abs(y)`.

The relationship between `floorDiv` and `floorMod` is such that:

- `floorDiv(x, y) * y + floorMod(x, y) == x`

See `Math.floorMod` for examples and a comparison to the `%` operator.

Parameters:

`x` - the dividend

`y` - the divisor

Returns:

the floor modulus `x - (floorDiv(x, y) * y)`

Throws:

`ArithmeticException` - if the divisor `y` is zero

Since:

1.8

See Also:

`Math.floorMod(int, int)`, `floorDiv(int, int)`

floorMod

```
public static long floorMod(long x,
                            long y)
```

Returns the floor modulus of the `long` arguments.

The floor modulus is `x - (floorDiv(x, y) * y)`, has the same sign as the divisor `y`, and is in the range of `-abs(y) < r < +abs(y)`.

The relationship between `floorDiv` and `floorMod` is such that:

- `floorDiv(x, y) * y + floorMod(x, y) == x`

See `Math.floorMod` for examples and a comparison to the `%` operator.

Parameters:

`x - the dividend`

`y - the divisor`

Returns:

`the floor modulus x - (floorDiv(x, y) * y)`

Throws:

`ArithmeticException - if the divisor y is zero`

Since:

`1.8`

See Also:

`Math.floorMod(long, long), floorDiv(long, long)`

abs

```
public static int abs(int a)
```

Returns the absolute value of an `int` value. If the argument is not negative, the argument is returned. If the argument is negative, the negation of the argument is returned.

Note that if the argument is equal to the value of `Integer.MIN_VALUE`, the most negative representable `int` value, the result is that same value, which is negative.

Parameters:

`a - the argument whose absolute value is to be determined.`

Returns:

`the absolute value of the argument.`

abs

```
public static long abs(long a)
```

Returns the absolute value of a `long` value. If the argument is not negative, the argument is returned. If the argument is negative, the negation of the argument is returned.

Note that if the argument is equal to the value of `Long.MIN_VALUE`, the most negative representable `long` value, the result is that same value, which is negative.

Parameters:

`a - the argument whose absolute value is to be determined.`

Returns:

`the absolute value of the argument.`

abs

```
public static float abs(float a)
```

Returns the absolute value of a `float` value. If the argument is not negative, the argument is returned. If the argument is negative, the negation of the argument is returned. Special cases:

- If the argument is positive zero or negative zero, the result is positive zero.
- If the argument is infinite, the result is positive infinity.
- If the argument is NaN, the result is NaN.

In other words, the result is the same as the value of the expression:

```
Float.intBitsToFloat(0x7fffffff & Float.floatToIntBits(a))
```

Parameters:

`a - the argument whose absolute value is to be determined`

Returns:

`the absolute value of the argument.`

abs

```
public static double abs(double a)
```

Returns the absolute value of a `double` value. If the argument is not negative, the argument is returned. If the argument is negative, the negation of the argument is returned. Special cases:

- If the argument is positive zero or negative zero, the result is positive zero.
- If the argument is infinite, the result is positive infinity.
- If the argument is NaN, the result is NaN.

In other words, the result is the same as the value of the expression:

```
Double.longBitsToDouble((Double.doubleToLongBits(a)<<1)>>>1)
```

Parameters:

`a - the argument whose absolute value is to be determined`

Returns:

the absolute value of the argument.

max

```
public static int max(int a,
                      int b)
```

Returns the greater of two `int` values. That is, the result is the argument closer to the value of `Integer.MAX_VALUE`. If the arguments have the same value, the result is that same value.

Parameters:

a - an argument.

b - another argument.

Returns:

the larger of a and b.

max

```
public static long max(long a,
                       long b)
```

Returns the greater of two `long` values. That is, the result is the argument closer to the value of `Long.MAX_VALUE`. If the arguments have the same value, the result is that same value.

Parameters:

a - an argument.

b - another argument.

Returns:

the larger of a and b.

max

```
public static float max(float a,
                        float b)
```

Returns the greater of two `float` values. That is, the result is the argument closer to positive infinity. If the arguments have the same value, the result is that same value. If either value is NaN, then the result is NaN. Unlike the numerical comparison operators, this method considers negative zero to be strictly smaller than positive zero. If one argument is positive zero and the other negative zero, the result is positive zero.

Parameters:

a - an argument.

b - another argument.

Returns:

the larger of a and b.

max

```
public static double max(double a,
                         double b)
```

Returns the greater of two `double` values. That is, the result is the argument closer to positive infinity. If the arguments have the same value, the result is that same value. If either value is NaN, then the result is NaN. Unlike the numerical comparison operators, this method considers negative zero to be strictly smaller than positive zero. If one argument is positive zero and the other negative zero, the result is positive zero.

Parameters:

a - an argument.

b - another argument.

Returns:

the larger of a and b.

min

```
public static int min(int a,
                      int b)
```

Returns the smaller of two `int` values. That is, the result the argument closer to the value of `Integer.MIN_VALUE`. If the arguments have the same value, the result is that same value.

Parameters:

a - an argument.

b - another argument.

Returns:

the smaller of a and b.

min

```
public static long min(long a,
                       long b)
```

Returns the smaller of two `long` values. That is, the result is the argument closer to the value of `Long.MIN_VALUE`. If the arguments have the same value, the result is that same value.

Parameters:

`a - an argument.`

`b - another argument.`

Returns:

`the smaller of a and b.`

min

```
public static float min(float a,
                        float b)
```

Returns the smaller of two `float` values. That is, the result is the value closer to negative infinity. If the arguments have the same value, the result is that same value. If either value is NaN, then the result is NaN. Unlike the numerical comparison operators, this method considers negative zero to be strictly smaller than positive zero. If one argument is positive zero and the other is negative zero, the result is negative zero.

Parameters:

`a - an argument.`

`b - another argument.`

Returns:

`the smaller of a and b.`

min

```
public static double min(double a,
                         double b)
```

Returns the smaller of two `double` values. That is, the result is the value closer to negative infinity. If the arguments have the same value, the result is that same value. If either value is NaN, then the result is NaN. Unlike the numerical comparison operators, this method considers negative zero to be strictly smaller than positive zero. If one argument is positive zero and the other is negative zero, the result is negative zero.

Parameters:

`a - an argument.`

`b - another argument.`

Returns:

`the smaller of a and b.`

ulp

```
public static double ulp(double d)
```

Returns the size of an ulp of the argument. An ulp, unit in the last place, of a `double` value is the positive distance between this floating-point value and the `double` value next larger in magnitude. Note that for non-NaN x, `ulp(-`x`) == ulp(`x`)`.

Special Cases:

- If the argument is NaN, then the result is NaN.
- If the argument is positive or negative infinity, then the result is positive infinity.
- If the argument is positive or negative zero, then the result is `Double.MIN_VALUE`.
- If the argument is ±`Double.MAX_VALUE`, then the result is equal to 2^{971}.

Parameters:

`d - the floating-point value whose ulp is to be returned`

Returns:

`the size of an ulp of the argument`

Since:

`1.5`

ulp

```
public static float ulp(float f)
```

Returns the size of an ulp of the argument. An ulp, unit in the last place, of a `float` value is the positive distance between this floating-point value and the `float` value next larger in magnitude. Note that for non-NaN x, `ulp(-`x`) == ulp(`x`)`.

Special Cases:

- If the argument is NaN, then the result is NaN.
- If the argument is positive or negative infinity, then the result is positive infinity.

- If the argument is positive or negative zero, then the result is `Float.MIN_VALUE`.
- If the argument is ±`Float.MAX_VALUE`, then the result is equal to 2^{104}.

Parameters:

`f - the floating-point value whose ulp is to be returned`

Returns:

`the size of an ulp of the argument`

Since:

`1.5`

signum

```
public static double signum(double d)
```

Returns the signum function of the argument; zero if the argument is zero, 1.0 if the argument is greater than zero, -1.0 if the argument is less than zero.

Special Cases:

- If the argument is NaN, then the result is NaN.
- If the argument is positive zero or negative zero, then the result is the same as the argument.

Parameters:

`d - the floating-point value whose signum is to be returned`

Returns:

`the signum function of the argument`

Since:

`1.5`

signum

```
public static float signum(float f)
```

Returns the signum function of the argument; zero if the argument is zero, 1.0f if the argument is greater than zero, -1.0f if the argument is less than zero.

Special Cases:

- If the argument is NaN, then the result is NaN.
- If the argument is positive zero or negative zero, then the result is the same as the argument.

Parameters:

`f - the floating-point value whose signum is to be returned`

Returns:

`the signum function of the argument`

Since:

`1.5`

sinh

```
public static double sinh(double x)
```

Returns the hyperbolic sine of a `double` value. The hyperbolic sine of *x* is defined to be $(e^x - e^{-x})/2$ where *e* is Euler's number.

Special cases:

- If the argument is NaN, then the result is NaN.
- If the argument is infinite, then the result is an infinity with the same sign as the argument.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`x - The number whose hyperbolic sine is to be returned.`

Returns:

`The hyperbolic sine of x.`

Since:

`1.5`

cosh

```
public static double cosh(double x)
```

Returns the hyperbolic cosine of a `double` value. The hyperbolic cosine of *x* is defined to be $(e^x + e^{-x})/2$ where *e* is Euler's number.

Special cases:

- If the argument is NaN, then the result is NaN.
- If the argument is infinite, then the result is positive infinity.
- If the argument is zero, then the result is `1.0`.

Parameters:

`x - The number whose hyperbolic cosine is to be returned.`

Returns:

The hyperbolic cosine of x.

Since:

1.5

tanh

```
public static double tanh(double x)
```

Returns the hyperbolic tangent of a `double` value. The hyperbolic tangent of *x* is defined to be $(e^x - e^{-x})/(e^x + e^{-x})$, in other words, sinh(x)/cosh(x). Note that the absolute value of the exact tanh is always less than 1.

Special cases:

- If the argument is NaN, then the result is NaN.
- If the argument is zero, then the result is a zero with the same sign as the argument.
- If the argument is positive infinity, then the result is `+1.0`.
- If the argument is negative infinity, then the result is `-1.0`.

Parameters:

`x - The number whose hyperbolic tangent is to be returned.`

Returns:

The hyperbolic tangent of x.

Since:

1.5

hypot

```
public static double hypot(double x,
                           double y)
```

Returns sqrt($x^2 + y^2$) without intermediate overflow or underflow.

Special cases:

- If either argument is infinite, then the result is positive infinity.
- If either argument is NaN and neither argument is infinite, then the result is NaN.

Parameters:

`x - a value`

`y - a value`

Returns:

sqrt($x^2 + y^2$) without intermediate overflow or underflow

Since:

1.5

expm1

```
public static double expm1(double x)
```

Returns e^x -1. Note that for values of x near 0, the exact sum of `expm1(x) + 1` is much closer to the true result of e^x than `exp(x)`.

Special cases:

- If the argument is NaN, the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is negative infinity, then the result is -1.0.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`x - the exponent to raise e to in the computation of e^x -1.`

Returns:

`the value e^x - 1.`

Since:

1.5

log1p

```
public static double log1p(double x)
```

Returns the natural logarithm of the sum of the argument and 1. Note that for small values `x`, the result of `log1p(x)` is much closer to the true result of ln(1 + `x`) than the floating-point evaluation of `log(1.0+x)`.

Special cases:

- If the argument is NaN or less than -1, then the result is NaN.
- If the argument is positive infinity, then the result is positive infinity.
- If the argument is negative one, then the result is negative infinity.
- If the argument is zero, then the result is a zero with the same sign as the argument.

Parameters:

`x - a value`

Returns:

`the value ln(x + 1), the natural log of x + 1`

Since:

1.5

copySign

```
public static double copySign(double magnitude,
                              double sign)
```

Returns the first floating-point argument with the sign of the second floating-point argument. For this method, a NaN `sign` argument is always treated as if it were positive.

Parameters:

`magnitude` - the parameter providing the magnitude of the result

`sign` - the parameter providing the sign of the result

Returns:

a value with the magnitude of magnitude and the sign of sign.

Since:

1.6

copySign

```
public static float copySign(float magnitude,
                             float sign)
```

Returns the first floating-point argument with the sign of the second floating-point argument. For this method, a NaN `sign` argument is always treated as if it were positive.

Parameters:

`magnitude` - the parameter providing the magnitude of the result

`sign` - the parameter providing the sign of the result

Returns:

a value with the magnitude of magnitude and the sign of sign.

Since:

1.6

getExponent

```
public static int getExponent(float f)
```

Returns the unbiased exponent used in the representation of a `float`. Special cases:

- If the argument is NaN or infinite, then the result is `Float.MAX_EXPONENT` + 1.
- If the argument is zero or subnormal, then the result is `Float.MIN_EXPONENT` -1.

Parameters:

`f - a float value`

Returns:

`the unbiased exponent of the argument`

Since:

`1.6`

getExponent

```
public static int getExponent(double d)
```

Returns the unbiased exponent used in the representation of a `double`. Special cases:

- If the argument is NaN or infinite, then the result is `Double.MAX_EXPONENT` + 1.
- If the argument is zero or subnormal, then the result is `Double.MIN_EXPONENT` -1.

Parameters:

`d - a double value`

Returns:

`the unbiased exponent of the argument`

Since:

`1.6`

nextAfter

```
public static double nextAfter(double start,
                               double direction)
```

Returns the floating-point number adjacent to the first argument in the direction of the second argument. If both arguments compare as equal the second argument is returned.

Special cases:

- If either argument is a NaN, then NaN is returned.

- If both arguments are signed zeros, `direction` is returned unchanged (as implied by the requirement of returning the second argument if the arguments compare as equal).
- If `start` is ±`Double.MIN_VALUE` and `direction` has a value such that the result should have a smaller magnitude, then a zero with the same sign as `start` is returned.
- If `start` is infinite and `direction` has a value such that the result should have a smaller magnitude, `Double.MAX_VALUE` with the same sign as `start` is returned.
- If `start` is equal to ± `Double.MAX_VALUE` and `direction` has a value such that the result should have a larger magnitude, an infinity with same sign as `start` is returned.

Parameters:

`start - starting floating-point value`

`direction - value indicating which of start's neighbors or start should be returned`

Returns:

`The floating-point number adjacent to start in the direction of direction.`

Since:

`1.6`

nextAfter

```
public static float nextAfter(float start,
                              double direction)
```

Returns the floating-point number adjacent to the first argument in the direction of the second argument. If both arguments compare as equal a value equivalent to the second argument is returned.

Special cases:

- If either argument is a NaN, then NaN is returned.
- If both arguments are signed zeros, a value equivalent to `direction` is returned.
- If `start` is ±`Float.MIN_VALUE` and `direction` has a value such that the result should have a smaller magnitude, then a zero with the same sign as `start` is returned.
- If `start` is infinite and `direction` has a value such that the result should have a smaller magnitude, `Float.MAX_VALUE` with the same sign as `start` is returned.
- If `start` is equal to ± `Float.MAX_VALUE` and `direction` has a value such that the result should have a larger magnitude, an infinity with same sign as `start` is returned.

Parameters:

`start - starting floating-point value`

`direction - value indicating which of start's neighbors or start should be returned`

Returns:

`The floating-point number adjacent to start in the direction of direction.`

Since:

`1.6`

nextUp

```
public static double nextUp(double d)
```

Returns the floating-point value adjacent to `d` in the direction of positive infinity. This method is semantically equivalent to `nextAfter(d, Double.POSITIVE_INFINITY);` however, a `nextUp` implementation may run faster than its equivalent `nextAfter` call.

Special Cases:

- If the argument is NaN, the result is NaN.
- If the argument is positive infinity, the result is positive infinity.
- If the argument is zero, the result is `Double.MIN_VALUE`

Parameters:

`d - starting floating-point value`

Returns:

`The adjacent floating-point value closer to positive infinity.`

Since:

`1.6`

nextUp

```
public static float nextUp(float f)
```

Returns the floating-point value adjacent to `f` in the direction of positive infinity. This method is semantically equivalent to `nextAfter(f, Float.POSITIVE_INFINITY);` however, a `nextUp` implementation may run faster than its equivalent `nextAfter` call.

Special Cases:

- If the argument is NaN, the result is NaN.
- If the argument is positive infinity, the result is positive infinity.

- If the argument is zero, the result is `Float.MIN_VALUE`

Parameters:

`f - starting floating-point value`

Returns:

`The adjacent floating-point value closer to positive infinity.`

Since:

`1.6`

nextDown

```
public static double nextDown(double d)
```

Returns the floating-point value adjacent to `d` in the direction of negative infinity. This method is semantically equivalent to `nextAfter(d, Double.NEGATIVE_INFINITY)`; however, a `nextDown` implementation may run faster than its equivalent `nextAfter` call.

Special Cases:

- If the argument is NaN, the result is NaN.
- If the argument is negative infinity, the result is negative infinity.
- If the argument is zero, the result is `-Double.MIN_VALUE`

Parameters:

`d - starting floating-point value`

Returns:

`The adjacent floating-point value closer to negative infinity.`

Since:

`1.8`

nextDown

```
public static float nextDown(float f)
```

Returns the floating-point value adjacent to `f` in the direction of negative infinity. This method is semantically equivalent to `nextAfter(f, Float.NEGATIVE_INFINITY)`; however, a `nextDown` implementation may run faster than its equivalent `nextAfter` call.

Special Cases:

- If the argument is NaN, the result is NaN.
- If the argument is negative infinity, the result is negative infinity.
- If the argument is zero, the result is `-Float.MIN_VALUE`

Parameters:

`f - starting floating-point value`

Returns:

`The adjacent floating-point value closer to negative infinity.`

Since:

`1.8`

scalb

```
public static double scalb(double d,
                           int scaleFactor)
```

Returns `d` × $2^{\texttt{scaleFactor}}$ rounded as if performed by a single correctly rounded floating-point multiply to a member of the double value set. See the Java Language Specification for a discussion of floating-point value sets. If the exponent of the result is between `Double.MIN_EXPONENT` and `Double.MAX_EXPONENT`, the answer is calculated exactly. If the exponent of the result would be larger than `Double.MAX_EXPONENT`, an infinity is returned. Note that if the result is subnormal, precision may be lost; that is, when `scalb(x, n)` is subnormal, `scalb(scalb(x, n), -n)` may not equal *x*. When the result is non-NaN, the result has the same sign as `d`.

Special cases:

- If the first argument is NaN, NaN is returned.
- If the first argument is infinite, then an infinity of the same sign is returned.
- If the first argument is zero, then a zero of the same sign is returned.

Parameters:

`d - number to be scaled by a power of two.`

`scaleFactor - power of 2 used to scale d`

Returns:

`d` × $2^{\texttt{scaleFactor}}$

Since:

`1.6`

scalb

```
public static float scalb(float f,
                          int scaleFactor)
```

Returns `f` × $2^{\texttt{scaleFactor}}$ rounded as if performed by a single correctly rounded floating-point multiply to a member of the float value set. See the

Java Language Specification for a discussion of floating-point value sets. If the exponent of the result is between `Float.MIN_EXPONENT` and `Float.MAX_EXPONENT`, the answer is calculated exactly. If the exponent of the result would be larger than `Float.MAX_EXPONENT`, an infinity is returned. Note that if the result is subnormal, precision may be lost; that is, when `scalb(x, n)` is subnormal, `scalb(scalb(x, n), -n)` may not equal *x*. When the result is non-NaN, the result has the same sign as `f`.

Special cases:

- If the first argument is NaN, NaN is returned.
- If the first argument is infinite, then an infinity of the same sign is returned.
- If the first argument is zero, then a zero of the same sign is returned.

Parameters:

`f - number to be scaled by a power of two.`

`scaleFactor - power of 2 used to scale f`

Returns:

$f \times 2^{scaleFactor}$

Since:

1.6

Submit a bug or feature
For further API reference and developer documentation, see Java SE Documentation. That documentation contains more detailed, developer-targeted descriptions, with conceptual overviews, definitions of terms, workarounds, and working code examples.
Copyright © 1993, 2026, Oracle and/or its affiliates. All rights reserved. Use is subject to license terms.
Also see the documentation redistribution policy.

Scripting on this page tracks web page traffic, but does not change the content in any way.